Public: Equities Overview

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1 Introduction

This document details on asset specific data infrastructure provided by MASSIVE.COM and its error modes and practical resolutions specifically to handle only diversified U.S. equity sector based mid frequency fundamental value to market value ratio forecasting.

2 Instruments and their Identifiers

Any tradable asset or financial contract (e.g., stock, ETF, or proxy) represented within the data environment as an **Instrument**. The term **assets** is too broad to describe the subset of items of the study, for example while Treasury Bonds could be traded, and certain etfs or international stocks can be traded either directly or indirectly, they are not stocks as they do not represent a share of a corporate entity. Thus the items that are traded on the Market Exchange using a name, which is their Ticker.

2.1 Ticker By Underlying Entity

An Instrument has a ticker, which is a hash reserved for the Instrument to be traded on the market. In other words a ticker and an instrument's relationship is analogous to a address and a person who lives there, a person could have many addresses over time, or could be addressless, but is strictly reserved to only having 1 address at any given time. A ticker and company name is chosen by the underlying company, which makes it a unreliable as it could be changed throughout time, see figure 1.

	id [PK] integer	composite_figi character varying (20)	composite_figi_hash bigint	ticker character varying (20)	ticker_hash bigint	name character varying (255)	cik bigint	sic_code integer	sic_description character varying (255)
9	16	BBG000B9XRY4	803323068916728167	AAPL	-6538189345531045678	APPLE INC	320193	3571	ELECTRONIC COMPUTERS
10	18	BBG000B9XRY4	803323068916728167	AAPL	-6538189345531045678	APPLE INC	320193	3571	ELECTRONIC COMPUTERS
11	20	BBG000B9XRY4	803323068916728167	AAPL	-6538189345531045678	Apple Inc.	320193	3571	ELECTRONIC COMPUTERS
12	22	BBG000B9XRY4	803323068916728167	AAPL	-6538189345531045678	Apple Inc.	320193	3571	ELECTRONIC COMPUTERS
13	24	BBG000B9XRY4	803323068916728167	AAPL	-6538189345531045678	Apple Inc.	320193	3571	ELECTRONIC COMPUTERS

Figure 1: Instruments, FIGI, Ticker, Name, CIK, SIC code, SIC Description

Thus names are assigned by third parties, which are more reliable, however this heavily increases reliance on the foreign entities. We focus on 3 other identifiers: Composite FIGI, CIK, and SIC code.

2.2 Financial Instrument Global Identifier (FIGI) by Bloomberg

Financial Instrument Global Identifier, also known as Composite FIGI, is created by Bloomberg to do quantitative analysis and other financial analysis, where the advantages are that it is very robust however there still exists edge cases when even Bloomberg is unable to label the data. See figure 2.

	id [PK] integer	composite_figi character varying (20)	composite_figi_hash bigint	ticker character varying (20)	ticker_hash bigint	name character varying (255)	cik bigint	sic_code integer	sic_description character varying (255)
25	1372	UNKNOWN	-8111899043101519345	AVGO	-1755704867092227164	Broadcom Limited Ordinary Shares	1649338	3674	SEMICONDUCTORS & RELATED DEVICES
26	1374	UNKNOWN	-8111899043101519345	AVGO	-1755704867092227164	Broadcom Limited Ordinary Shares	1649338	3674	SEMICONDUCTORS & RELATED DEVICES
27	1376	UNKNOWN	-8111899043101519345	AVGO	-1755704867092227164	Broadcom Limited Ordinary Shares	1649338	3674	SEMICONDUCTORS & RELATED DEVICES
28	1378	UNKNOWN	-8111899043101519345	AVGO	-1755704867092227164	Broadcom Limited Ordinary Shares	1649338	3674	SEMICONDUCTORS & RELATED DEVICES
29	1380	BBG00KHY5S69	989568436600796786	AVGO	-1755704867092227164	Broadcom Inc. Common Stock	1730168	3674	SEMICONDUCTORS & RELATED DEVICES
30	1382	BBG00KHY5S69	989568436600796786	AVGO	-1755704867092227164	Broadcom Inc. Common Stock	1730168	3674	SEMICONDUCTORS & RELATED DEVICES

Figure 2: Unknown FIGI on Broadcom (AVGO), where the Bloomberg Analysis Team did not cover an Instrument

2.3 Central Index Key (CIK) by SEC

CIK is the code created by the U.S. Securities and Exchange Commission (SEC) to track corporate and individual disclosure legal and formal filings. This is relatively robust, however this fails on cases that companies divide their shares by voting and nonvoting shares, that are both tradeable on the market with different tickers. While they both exhibit same prices and same accounting values (see figure 4), Bloomberg still classifies them as different.

	id [PK] integer	composite_figi character varying (20)	composite_figi_hash bigint	ticker character varying (20)	ticker_hash bigint	name character varying (255)	cik bigint	sic_code integer	sic_description character varying (255)
113	5876	BBG009S39JX6	7656398217336052926	GOOGL	-431372410334157488	Alphabet Inc. Class A Common S...	1652044	7370	SERVICES-COMPUTER PROGRAMMING, DATA PROCESSING, ETC.
114	5877	BBG009S3NB30	-3320472440211998197	GOOG	-3497053127626027877	Alphabet Inc. Class C Capital Stock	1652044	7370	SERVICES-COMPUTER PROGRAMMING, DATA PROCESSING, ETC.
115	5878	BBG009S39JX6	7656398217336052926	GOOGL	-431372410334157488	Alphabet Inc. Class A Common S...	1652044	7370	SERVICES-COMPUTER PROGRAMMING, DATA PROCESSING, ETC.
116	5879	BBG009S3NB30	-3320472440211998197	GOOG	-3497053127626027877	Alphabet Inc. Class C Capital Stock	1652044	7370	SERVICES-COMPUTER PROGRAMMING, DATA PROCESSING, ETC.
117	5880	BBG009S39JX6	7656398217336052926	GOOGL	-431372410334157488	Alphabet Inc. Class A Common S...	1652044	7370	SERVICES-COMPUTER PROGRAMMING, DATA PROCESSING, ETC.

Figure 3: Duplicate CIK, conflicting Indentification protocol with Bloomberg FIGI

	id [PK] bigint	ticker character varying (20)	ticker_hash bigint	filing_date date	period_end date	timeframe character varying (30)	basic_earnings_per_share numeric (20,2)	basic_shares_outstanding numeric (20,2)	consolidated_net_income_loss numeric (20,2)	cost_of_revenue numeric (20,2)
1	4264	GOOGL	-431372410334157488	2026-04-30	2026-03-31	quarterly	5.17	12099000000.00	62578000000.00	41271000000.00
2	8296	GOOG	-3497053127626027877	2026-04-30	2026-03-31	quarterly	5.17	12099000000.00	62578000000.00	41271000000.00
3	8320	GOOG	-3497053127626027877	2026-02-05	2025-12-31	quarterly	2.85	12116000000.00	34455000000.00	45767000000.00
4	4290	GOOGL	-431372410334157488	2026-02-05	2025-12-31	quarterly	2.85	12116000000.00	34455000000.00	45767000000.00
5	8339	GOOG	-3497053127626027877	2025-10-30	2025-09-30	quarterly	2.89	12086000000.00	34979000000.00	41369000000.00
6	4312	GOOGL	-431372410334157488	2025-10-30	2025-09-30	quarterly	2.89	12086000000.00	34979000000.00	41369000000.00
7	4334	GOOGL	-431372410334157488	2025-07-24	2025-06-30	quarterly	2.33	12122000000.00	28196000000.00	39039000000.00
8	8358	GOOG	-3497053127626027877	2025-07-24	2025-06-30	quarterly	2.33	12122000000.00	28196000000.00	39039000000.00

Figure 4: Alphabet Class C, A (GOOG, GOOGL), exhibit same accounting data

2.4 SIC Code by Interdepartmental Committee on Industrial Statistics

Although this is not a unique Identifier, the SIC code created by Interdepartmental Committee on Industrial Statistics is used to track economic statistics. It is a quantitative and hierarchal tag that determines a company's industry and specific business model. Furthermore, the SIC code has a semantic tag of what the quantitative hierarchal tag, called sic description, see figure 3. Although the Public sector has deemed SIC code deprecated and replaced with NAICS code, NAICS Code Endpoint, SIC code is quite popular within the private sector, as it is publically offered and supported by the Massive.com Data Endpoint. NAICS is out of scope as of 2026/07/29.

2.5 Other Information

The locale is the country of origin of the listed instrument. The city, postal code, state is where the company declares its headquarters to be located. Total employees is the approximated number of persons under payroll under a specific company. Shares outstanding is the number of shares that are currently on the market available to trade. The market cap is the Shares outstanding multiplied by the price of that company's stock.

The market as Massive.com Data Endpoint refers to the categorization of what exchange sector they are listed in: crypto, fx, indices, otc, stocks.

1. crypto: cryptocurrency exchange. eg. Bitcoin, Ethereum.
2. fx: Foreign Exchange. eg. USD/EUR (EUR=X) exchange rate.
3. indices: Indexes, cannot be traded by themselves, but can be traded through options.
4. otc: Pink Sheet stocks, small high risk companies, not approved to be listed in the NYSE yet, or ever, or delisted from Large Exchanges.
5. stocks: Large Exchanges including the subset of NYSE (New York, USA), TSX (Toronto, Canada), LSE (London, United Kingdom).

For example Bitcoin itself cannot be traded on the NYSE, and must be traded on a cryptocurrency market, similarly, GOOG cannot be traded on the crypto market. The primary exchange is the specific large exchange that an instrument is traded on, specifically NYSE, or TSX, or LSE. The metric **active** is a boolean flag indicating whether the instrument is remains an active data feed at a given point in time.

2.6 Instrument Type

While there are many different instrument types, only CS is relevant because only CS has accounting data to compute a fundamental valuation for a company. Due to this requirement to need a physical asset backed valuation, the other intrument types are practically useless to model as Massive.com Data Endpoint does not provide their accounting data for one to model an asset backed valuation.

Code	**Description**
CS	Common Stock (e.g., Apple)
ADRC	Common Foreign Stock (ADR)
ADRP	Preferred Foreign Stock (ADR)
UNIT	Bundled Security (Common + Warrant)
RIGHT	Short-term Subscription Rights
PFD	Preferred Stock (Fixed Dividend)
FUND	Mutual Fund or Closed-End Fund
WARRANT	Derivative Right to Buy Asset
ETF	Exchange Traded Fund (Basket)
ETN	Exchange Traded Note (Debt-based)
OS	Foreign Common Shares
GDR	Global Depositary Receipt
NYRS	New York Registry Shares (Dutch/German)
AGEN	Government Agency Debt
BOND	Fixed Income Debt
OTHER	Alternative Assets (e.g., Fractional Real Estate)

3 Many to Many Entity Mapping

To identify an entity is tricky, because of multiple parties assigning names with different protocols.

- **Ticker Swaps:** When a company changes its symbol (e.g., FB, META).
- **Absorptions/Mergers/Aquisitions:** When one entity is consumed by another.

- **IPOs:** The initial creation of the data record for a new entity.
- **Ticker Reuse:** When a dead symbol is reclaimed by a new, unrelated entity (e.g., FB being reused by an ETF).

Google was known as BackRub in 1994, yet you would never refer to Google as Backrub. Thus we need to root the name by the most recent name, and trace back. This is done literally, a sweep of stride (1 or 3 months) is done backwards in time with the most recent ticker to derive the point in time entity ticker, name, figi, with the parameter "asof time" or "point in time" and the parameter ticker.

3.1 API Failure

Unfortunately Massive.com Data Endpoint does not offer the relational database queries and thus a sweep with stride with a set of tickers is the only way to serve as a proxy to the relational point in time database. Furthermore their API that supposedly serves as solution to their inability to offer direct relational database connection serves faulty data.

For example, the asset **LCID** (Lucid Group, Inc.) shows a listing date of `2021-07-26`. However, a query to the ticker events endpoint returns a single ticker change dated `2025-09-02`, completely omitting the initial listing and any interim transitions:

Listing 1: Faulty API Response for Ticker LCID


```
{
  "results": {
    "name": "Lucid Group, Inc. Common Stock",
    "composite_figi": "BBG00W7F99V5",
    "events": [
      {
        "ticker_change": { "ticker": "LCID" },
        "type": "ticker_change",
        "date": "2025-09-02"
      }
    ]
  },
  "status": "OK"
}
```


Thus the proxy endpoint cannot be used as a source of truth for historical reconstruction. However the proxy stride solution is a robust workaround to this faulty endpoint.

A property can be derived, if the set of tickers (FB, META) share a FIGI and they share contiguous start and end strides, they must be the same entity, see figure 5. Furthermore, they must have swapped names sometime after the end of the last instance of the first name future-padded by the stride length, and sometime before the second name, past-padded by the stride length. This produces 2 edge cases where we cannot guarantee the exact date when ticker is swapped for all entities in reasonable time, and the second edge case that we must know all historical tickers that we want to model. The good news is that the second edge case is that ticker swaps are relatively rare, and furthermore if one doesn't know or remember or can't find the historical ticker, one cannot model it. In other words, one must need the ability to get it to model the US Bureacratic Financial System with minimal preventable edge cases.

The deduction is that none of FIGI, ticker, CIK, SIC code are all not unique, see section 2.1, 2.2, 2.3 , 2.4. Thus none of FIGI, ticker, CIK, SIC code alone is enough to map a instrument to its underlying corporation. Thus a multi-key is needed to deduce a entity, to refer to a entity you would need to know atleast, both its FIGI and ticker. Thus this produces a hypothetical

	composite_figi character varying (20)	ticker character varying (20)	name character varying (255)	cik bigint	point_in_time_date date
45	BBG000MM2P62	META	Meta Platforms, Inc. Class A Common Stock	1326801	2022-11-01
46	BBG000MM2P62	META	Meta Platforms, Inc. Class A Common Stock	1326801	2022-10-01
47	BBG000MM2P62	META	Meta Platforms, Inc. Class A Common Stock	1326801	2022-09-01
48	BBG000MM2P62	META	Meta Platforms, Inc. Class A Common Stock	1326801	2022-08-01
49	BBG000MM2P62	META	Meta Platforms, Inc. Class A Common Stock	1326801	2022-07-01
50	BBG000MM2P62	FB	Meta Platforms, Inc. Class A Common Stock	1326801	2022-06-01
51	BBG000MM2P62	FB	Meta Platforms, Inc. Class A Common Stock	1326801	2022-05-01
52	BBG000MM2P62	FB	Meta Platforms, Inc. Class A Common Stock	1326801	2022-04-01
53	BBG000MM2P62	FB	Meta Platforms, Inc. Class A Common Stock	1326801	2022-03-01

Figure 5: FB META mapping to same entity

edge cases where figi, ticker multi-key is not unique, but the solution is to include cik as a new 3 factor multi-key. But this hypothetical edge case is non-existent so far.

4 Stock Price and Stock Splits

4.1 Stock Price

A price is defined by an asset ticker and a timestamp. Since the goal is mid-frequency forecasting on a Modest (but Modern) High-Performance-Computer, tradeoff is needed to preserve precious RAM, SSD, and compute time. Thus 15,30 minute frequencies is the sweet spot between API access time and local storage. Although Ideally 1,2 hour frequencies would be sweeter, but the API endpoint needs to compute open-high-low-close on their server, which decreases download throughput. Although higher resolution is possible, such as 1,5 minute frequencies, it is smoothed anyway, as the intended algorithm is quite computationally expensive and high frequency price actions are quite noisy and susceptible to high frequency-Trading order-front running Tsay [2010], LoM [2008], while providing marginal signals.

Massive.com Data Endpoint provided Volume Weighted Average Price (vw, VWAP) (`vw`) field is based on the download parameters on the frequency intervals see figure 6

4.2 Stock Splits

A Company stock splits to increase liquidity of their shares and to attract new investors, or abide by new legal laws. At any given time, a stock has a adjusted price and unadjusted price, with adjusted price being the current price, with respect to current shares outstanding, and a historical price- the nominal price with respect to historical shares outstanding. By keeping track of both adjusted price and unadjusted price with a flag, you can deduce the split ratio, see Nvidia's (NVDA) stock split 10 to 1 in figure 6.

This is particularly useful because this yields a real-time ratio which we can deduce the post-split adjusted shares outstanding from the most recent quarterly income statement. While it is possible to deduce shares outstanding without the dual price stream, relying on quarterly financial reports has a approximate 45 day latency. By querying both the raw nominal price stream (P_t^{raw}) and the split-adjusted price stream (P_t^{adj}) simultaneously, the latent corporate action multiplier (M_t) is dynamically isolated on the fly:

$$M_t = \text{round}\left(\frac{P_t^{raw}}{P_t^{adj}}\right) \tag{1}$$

	ticker character varying (20)	ticker_hash bigint	t bigint	c numeric (18,4)	v numeric (24,10)	vw numeric (18,4)	adjusted boolean
29233	NVDA	3160405114793146702	1709634600000000000	84.7200	661040.0000000000	84.5264	true
29234	NVDA	3160405114793146702	1709634600000000000	847.2000	66104.0000000000	845.2641	false
29235	NVDA	3160405114793146702	1709632800000000000	842.0000	184333.0000000000	837.5456	false
29236	NVDA	3160405114793146702	1709632800000000000	84.2000	1843330.0000000000	83.7546	true
29237	NVDA	3160405114793146702	1709631000000000000	83.7100	1615190.0000000000	84.1663	true
29238	NVDA	3160405114793146702	1709631000000000000	837.1000	161519.0000000000	841.6628	false
29239	NVDA	3160405114793146702	1709629200000000000	851.8900	77670.0000000000	853.1045	false
29240	NVDA	3160405114793146702	1709629200000000000	85.1890	776700.0000000000	85.3105	true
29241	NVDA	3160405114793146702	1709598600000000000	85.7950	907410.0000000000	85.9017	true
29242	NVDA	3160405114793146702	1709598600000000000	857.9500	90741.0000000000	859.0174	false

Figure 6: NVDA's Stock 1 to 10 split

This multiplier M_t represents the exact structural scaling factor required to transition between nominal and split-adjusted coordinates at execution timestamp t.

5 Financial Statements

Filed Quarterly these statements are semantically derived from Massive.com Data Endpoint private proprietary scraping algorithm from the SEC's eXtensible Business Reporting Language (XBRL) files SEC Filing Endpoint. Use the Multi-Key Mapping to find these Financial statements. Financial Statements can be expressed in Quarterly Reports, Annual Reports, and Trailing Twelve Months (TTM). To account for seasonality, Institutions uses TTM which is a rolling 4 quarter Cumulative Sum. Unlike Income Statements and Cash Flow Statements, Balance Sheets is a snapshot in its inherent property, it does not have a TTM property. For example if one owns a car, one would not compute a rolling cumulative sum of the cars market value as their Net Worth, they would just compute the most recent car value. However if one works a seasonal job, one must need to compute a TTM income to estimate their expected income. Similarly, a company's assets is a snapshot, and their revenues and free cash flow would need a Trailing Twelve month value to account for seasonality.

Table 1: Financial Statement Variables

Table Name	**Variable Name**
Balance Sheet	accounts_payable
	accrued_and_other_current_liabilities
	accumulated_other_comprehensive_income
	additional_paid_in_capital
	cash_and_equivalents
	commitments_and_contingencies
	common_stock
	debt_current
	deferred_revenue_current
	goodwill
	intangible_assets_net
	inventories
	long_term_debt_and_capital_lease_obligations
	noncontrolling_interest
	other_assets
	other_current_assets

Table Name	Variable Name
	other_equity
	other_noncurrent_liabilities
	preferred_stock
	property_plant_equipment_net
	receivables
	retained_earnings_deficit
	short_term_investments
	total_assets
	total_current_assets
	total_current_liabilities
	total_equity
	total_equity_attributable_to_parent
	total_liabilities
	total_liabilities_and_equity
	treasury_stock
Cash Flow	cash_from_operating_activities_continuing_operations
	change_in_cash_and_equivalents
	change_in_other_operating_assets_and_liabilities_net
	depreciation_depletion_and_amortization
	dividends
	effect_of_currency_exchange_rate
	income_loss_from_discontinued_operations
	long_term_debt_issuances_repayments
	net_cash_from_financing_activities
	net_cash_from_financing_activities_continuing_operations
	net_cash_from_financing_activities_discontinued_operations
	net_cash_from_investing_activities
	net_cash_from_investing_activities_continuing_operations
	net_cash_from_investing_activities_discontinued_operations
	net_cash_from_operating_activities
	net_cash_from_operating_activities_discontinued_operations
	net_income
	noncontrolling_interests
	other_cash_adjustments
	other_financing_activities
	other_investing_activities
	other_operating_activities
	purchase_of_property_plant_and_equipment
	sale_of_property_plant_and_equipment
	short_term_debt_issuances_repayments
Income Statement	basic_earnings_per_share
	basic_shares_outstanding
	consolidated_net_income_loss
	cost_of_revenue
	depreciation_depletion_amortization
	diluted_earnings_per_share
	diluted_shares_outstanding
	discontinued_operations
	ebitda
	equity_in_affiliates

Table Name	Variable Name
	extraordinary_items
	gross_profit
	income_before_income_taxes
	income_taxes
	interest_expense
	interest_income
	net_income_loss_attributable_common_shareholders
	noncontrolling_interest
	operating_income
	other_income_expense
	other_operating_expenses
	preferred_stock_dividends_declared
	research_development
	revenue
	selling_general_administrative
	total_operating_expenses
	total_other_income_expense
Short Interest	settlement_date
	short_interest
	avg_daily_volume
	days_to_cover
Dividends	record_date
	pay_date
	declaration_date
	ex_dividend_date
	frequency
	cash_amount
	currency
	distribution_type
	ticker
	historical_adjustment_factor
	split_adjusted_cash_amount

5.1 Feature Space Dimensionality Reduction via Variance Regularization

Following initial ratio synthesis across the three statements, a disciplined pruning step is necessary to eliminate low-information-density series and enforce balanced variance across the input manifold. Directly feeding unpruned accounting vectors into gradient-based optimizers introduces severe numerical conditioning challenges: when features span disparate orders of dispersion, standard optimization algorithms suffer from gradient attenuation on low-magnitude dimensions.

A representative case of an uninformative dimension is `set4_equity_in_affiliates_ttm`, whose empirical cross-sectional statistics across $N = 2{,}203$ observations are summarized in Table 2.

Table 2: Empirical Distribution of a Candidate Pruned Series

Feature Metric	n	μ	Median	σ	Min	Max
`set4_equity_in_affiliates_ttm`	2,203	0.0001	0.0000	0.0015	-0.0194	0.0236

This exclusion does not imply that unconsolidated affiliate earnings lack economic relevance in corporate accounting. Rather, the decision is dictated by representation mechanics and scaling constraints:

1. **Extreme Sparsity and Zero-Entropy Mass:** With an empirical median of 0.0000 and a standard deviation of $\sigma = 0.0015$, the overwhelming majority of issuers report zero affiliate activity under the equity method. The feature functions effectively as a dead channel with near-zero mutual information relative to broader cross-sectional market premia.

2. **Failure Modes of Independent Scaling:** Retaining such low-variance series necessitates non-uniform transformation regimes:

   - *Ad-Hoc Scaling Factors:* Scaling low-dispersion features by arbitrary constants destroys structural comparability across common-sized components within the same statement family.
   - *Standardization Drift ($\mathcal{Z}$-Scoring):* Applying dynamic time-series standardization (e.g., rolling standard scaling) injects lookahead artifacts, non-stationary variance shifts, and temporal instability across sliding windows, degrading the point-in-time integrity of the tensor.

3. **Downstream Absorption:** Because affiliate earnings are recognized downstream on the income statement, their economic contribution is preserved in consolidated bottom-line figures (EBT_{ttm} and NI_{ttm}), mitigating the risk of structural information loss upon removal.

Consequently, candidate dimensions exhibiting empirical dispersion thresholds below critical bounds ($\sigma < \tau_\sigma$) coupled with structural sparsity ($\mathbb{P}(x = 0) > 0.95$) are systematically pruned. This conservative filtering reduces parameter overhead, eliminates degenerate dimensions, and guarantees that the surviving feature space maintains balanced numerical bounds across the full training universe.

5.2 Short Interest and Dividend Variables

Short interest data is updated on a biweekly reporting cadence and tracks market-wide positioning against a given security. The `settlement_date` marks the official reporting cutoff

date for exchange-reported positions. The total volume of shares borrowed and held short by investors is recorded in `short_interest`. To contextualize position size relative to market liquidity, `avg_daily_volume` measures the average trading volume over the preceding period, which directly calculates `days_to_cover`—the ratio of short interest to average daily volume representing the estimated time required for short sellers to buy back and cover their positions.

Dividend metrics are typically announced and distributed on a quarterly schedule. Key timeline milestones include `declaration_date` (when corporate board approval is announced), `ex_dividend_date` (the cutoff date determining share ownership eligibility for the payout), `record_date` (when ownership rolls are finalized), and `pay_date` (when cash distributions are disbursed). The `frequency` parameter indicates payment intervals per year, while `cash_amount` and `currency` detail the unadjusted per-share payout and its unit denomination. The corporate action classification is specified by `distribution_type` (e.g., ordinary versus special dividends), and `ticker` identifies the corresponding asset. To maintain consistent historical time series, `historical_adjustment_factor` captures corporate stock split adjustments, enabling the calculation of `split_adjusted_cash_amount` to reflect real per-share cash returns accurately over time.

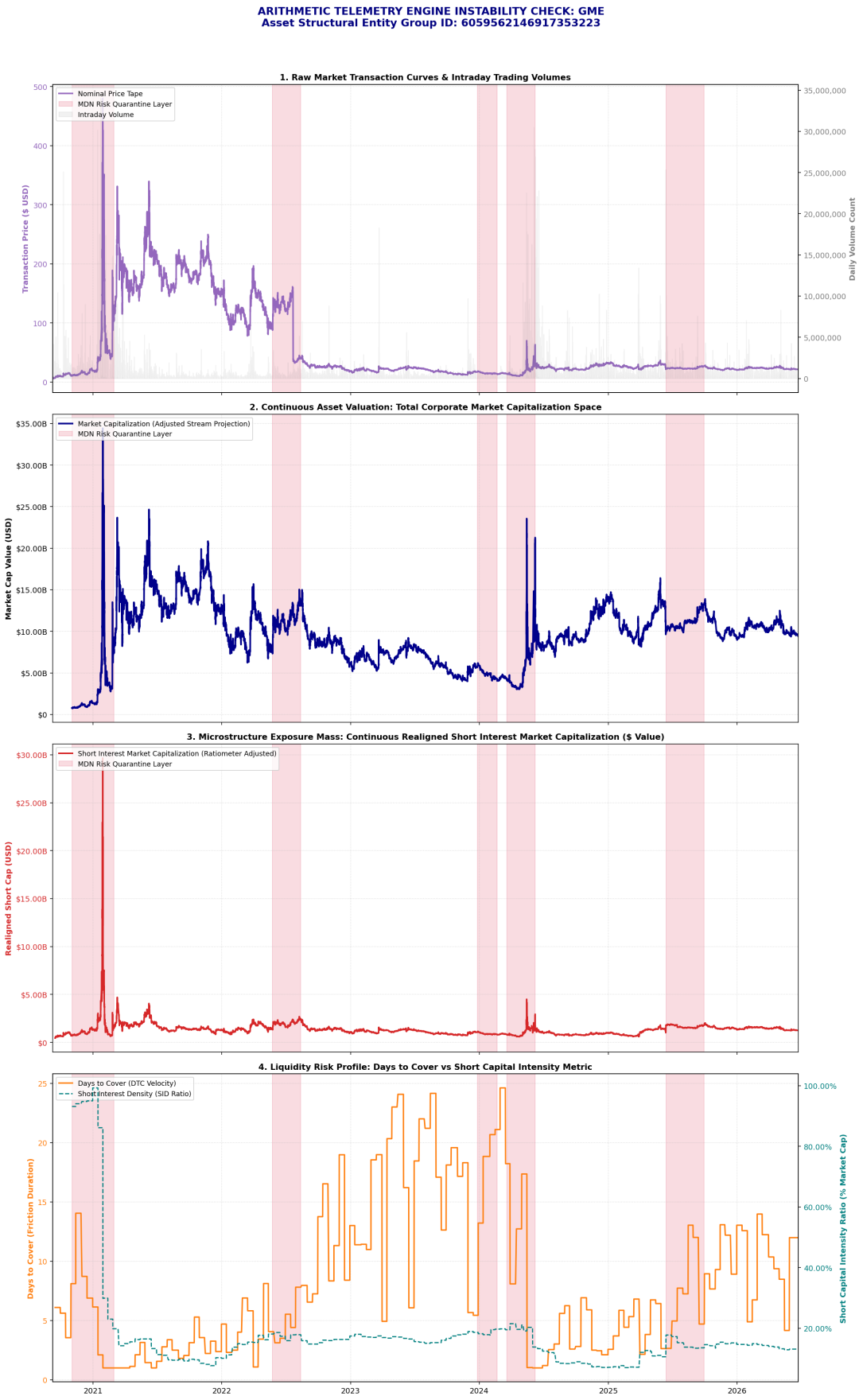


Figure 7: Arithmetic telemetry profile for GameStop (GME).

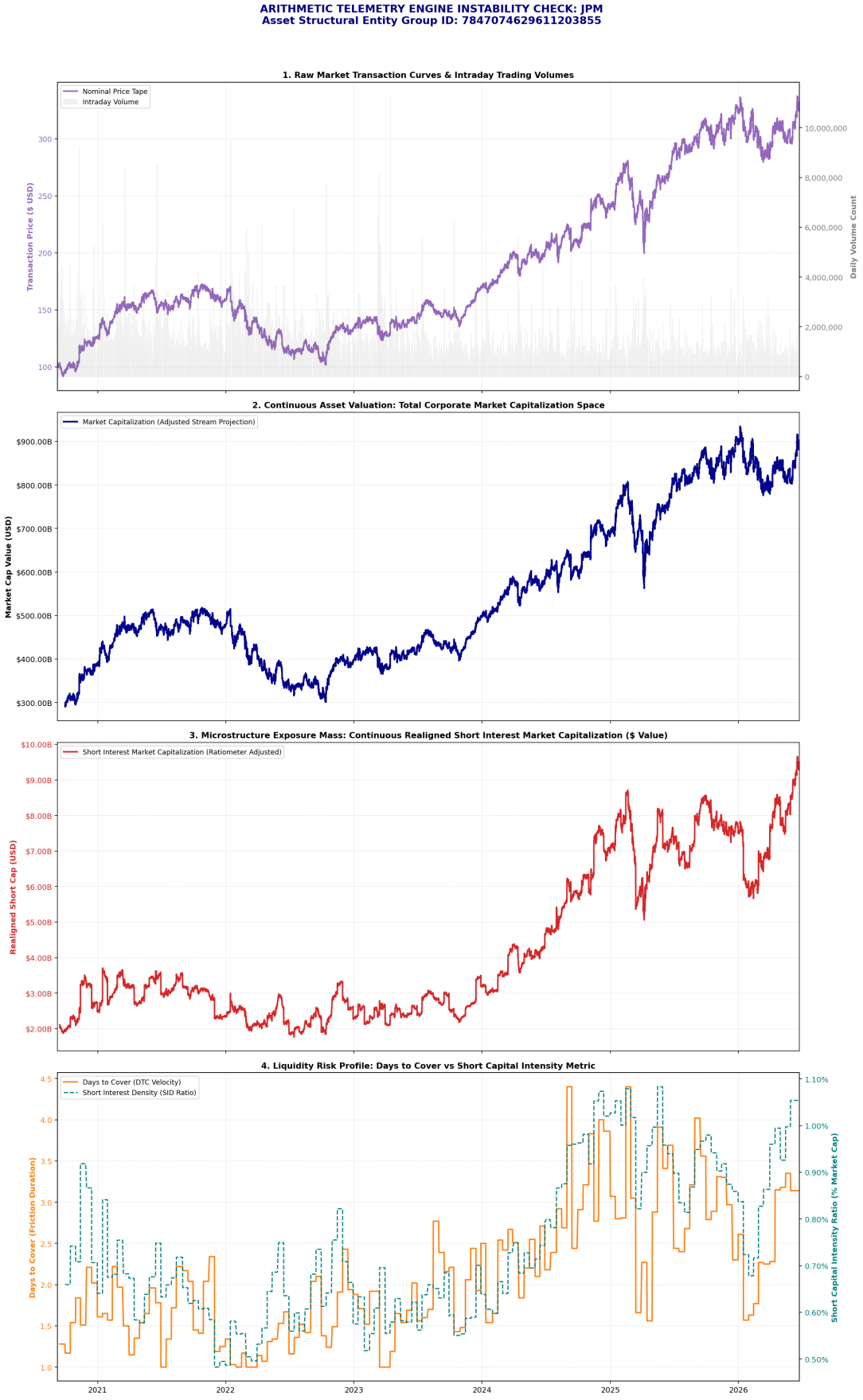


Figure 8: Arithmetic telemetry profile for JPMorgan Chase (JPM).

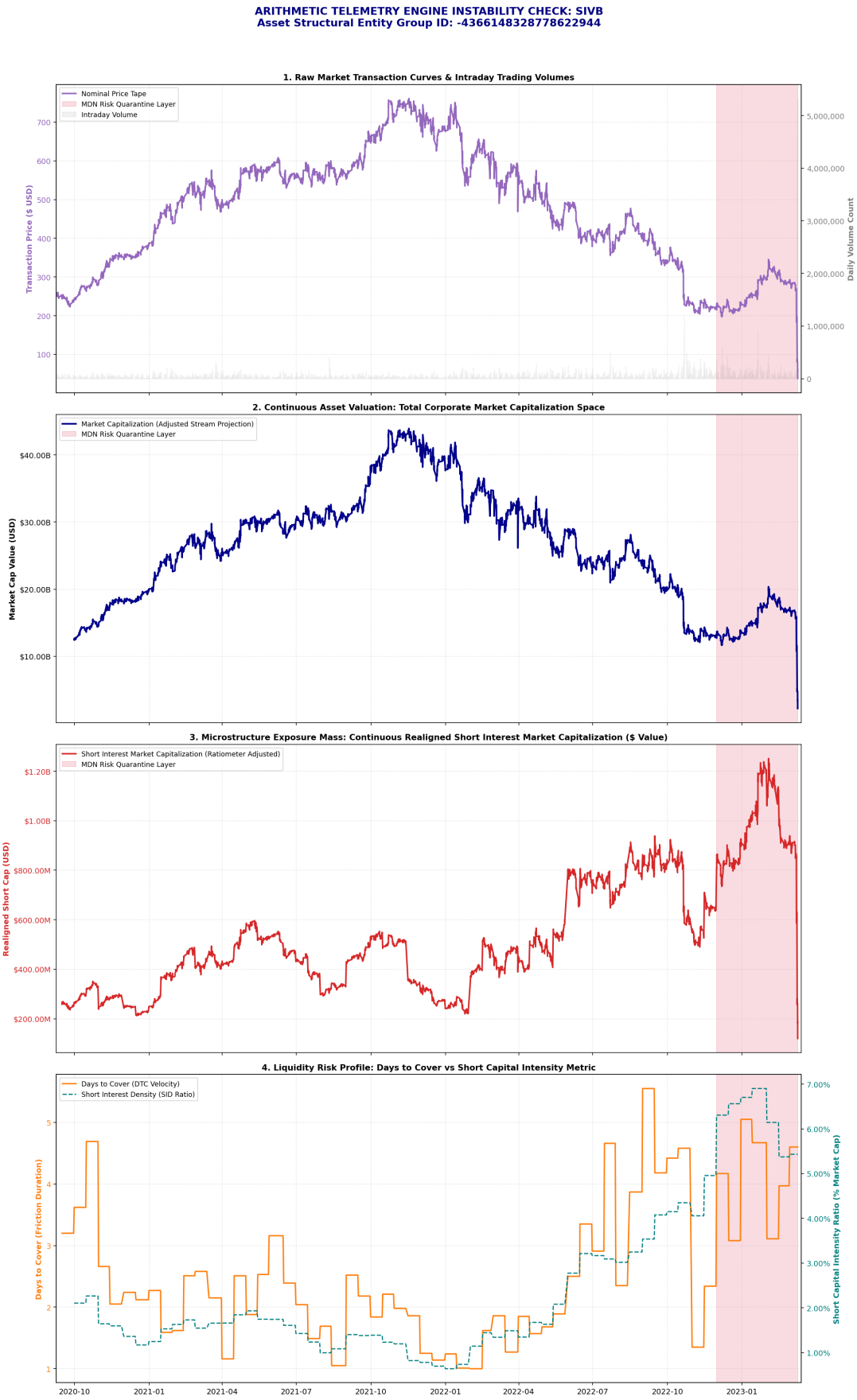


Figure 9: Arithmetic telemetry profile for Silicon Valley Bank (SIVB).

6 Fundamental Valuation

Stock Prices as we understand them asof 2026/07/29: Stock prices must satisfy the following three fundamental properties:

1. **Bounded Determinism:** Prices are structurally bounded and non-arbitrary.
2. **Martingale Departure:** Price trajectories systematically diverge from pure martingale processes.
3. **Behavioral Non-Linearity:** Price discovery exhibits stochastic anomalies and non-rational mechanics.

6.1 Ad Hoc Derivation of Bounded, Non-Arbitrary Pricing

To establish that market valuations are structurally bounded, we examine a nonrigous rhetorical boundary conditions across enterprise scales:

1. **Upper Bound (Base Case):** Consider a enterprise initialized with zero capital assets (spanning physical, human, intellectual, and infrastructure capital) and zero revenue. The value cannot exceed zero.
2. **Lower Bound (Base Case):** A company that gets listed on the public exchange cannot drop below zero ($P \geq 0$).
3. **Upper Bound (Scaling Step):** Extending this framework to mega-cap equities—such as NVIDIA Corporation ($\approx$ \$5 Trillion market capitalization as of July 29, 2026, with TTM revenue of \$253.49 Billion and total assets of \$259.47 Billion for the period ending April 26, 2026)—market values remain economically constrained. An arbitrary valuation (e.g., \$1 Quadrillion) is bounded above by global macroeconomic capital and liquidity constraints.
4. **Lower Bound (Scaling Step):** Conversely, NVIDIA's market valuation statistically unlikely fall below the aggregate liquidation value of its immediate net assets, given historically relative market premium ratio. Utilizing financial metrics across the balance sheet, cash flow, and income statements, it is theoretically possible to establish a strict fundamental valuation floor Koller et al. [2021], but it is practically impossible to do in practice (see **??** , **??**, **??**, **??**, **??**, **??**).

By induction across enterprise scales, all equities possess a non-arbitrary, fundamental distressed valuation floor above which market prices must trade. This structural lower bound aligns directly with Merton's option-pricing theory, wherein equity is modeled as a European call option on total firm asset value with a strike price equal to nominal debt obligations Merton [1974].

6.2 Structural Failure of the Pure Martingale Assumption

Samuelson's seminal formulation posits that under full informational efficiency, properly anticipated asset prices follow a martingale process Samuelson [1973]. However, real-world financial systems contradict pure martingale behavior due to non-linear feedback mechanisms inherent to complex systems: expanding revenues build investor confidence, which expands capital access, accelerates expenditure, and further boosts revenue.

Conversely, endogenous feedback loops frequently induce asset bubbles and systemic crashes. If the Efficient Market Hypothesis (EMH) Fama [1970] held under a strict martingale process, asset bubbles would not systematically form and burst; assuming prices are unforecastable

martingales often begs the question by assuming a state of perfect efficiency without accounting for market feedback loops.

6.3 Empirical Evidence of Non-Rational Price Dynamics

Whilst this section is evidence of survivorship bias, it does not take away the validity of the property. Market dynamics frequently demonstrate structural deviations from rational price discovery. A primary illustration is GameStop (GME) see figure 7, where short interest rose to extreme proportions relative to historical market norms (where short interest typically averages $\approx 1\%$ of float. see figure 8), indicating irrational crowding and market positioning.

Similarly, Silicon Valley Bank (SIVB) see figure 9 experienced substantial increases in short interest months prior to its insolvency, signaling that informed market participants detected severe structural vulnerabilities. Despite these public signals, the stock price remained largely stagnant before undergoing a sudden, catastrophic $\approx 80\%$ collapse over a two-day window. This temporal lag highlights a failure of continuous price discovery, as market prices failed to incorporate short-side informational signals until systemic liquidity collapsed.

6.4 Dividend-Induced Contango and Backwardation

In the context of equity derivatives (specifically Stock Futures), the relationship between the **Spot Price** and the **Futures Price** is fundamentally driven by the Cost of Carry. Dividends are the most critical variable in this equation, often flipping the market between Contango and Backwardation Fama [1970], Merton [1974].

6.4.1 The Mathematical Anchor: Cost of Carry

The fair value of a stock future is determined by the interest cost to hold the stock minus the expected dividend yield:

$$F = S \times e^{(r-q)T} \tag{2}$$

Where:

- F = Futures Price
- S = Spot (Current) Price
- r = Risk-free interest rate
- q = Dividend Yield
- T = Time to maturity

6.4.2 Dividend-Driven Backwardation ($q > r$)

Backwardation occurs when the Futures Price trades at a discount to the Spot Price ($F < S$). In equity markets, this structural discount is almost exclusively driven by an upcoming significant dividend distribution.

- **The Logic:** Holders of physical spot shares receive the corporate cash dividend, whereas long futures contract holders do not.
- **The Adjustment:** To compensate futures holders for missing the cash payout, the derivative market prices the future at a discrete discount relative to spot.
- **Market State:** When a large-cap constituent announces a substantial dividend yield, stock futures trade below spot until the ex-dividend date passes.

6.4.3 Dividend-Driven Contango ($r > q$)

Contango occurs when the Futures Price trades at a premium to the Spot Price ($F > S$). This represents the baseline state for non-dividend-paying equities or assets with minimal yields.

- **The Logic:** Purchasing physical spot equity upfront requires capital allocation that would otherwise earn the risk-free rate (r).
- **The Adjustment:** The futures contract trades at a premium to reflect the cost of financing and carrying the underlying asset position over time T.
- **Market State:** For non-dividend-paying equities, the futures price sits continuously above spot, slowly converging downward toward the spot price as expiration approaches.

6.4.4 Ex-Dividend Convergence Mechanics

Regardless of whether the market regime is in contango or backwardation, a structural realignment occurs on the **Ex-Dividend Date**:

1. **Spot Price Reset:** On the morning of the ex-dividend date, the spot price undergoes an immediate downward adjustment reflecting the net cash outflow from firm assets.
2. **Basis Convergence:** Because the pending dividend event has passed and is stripped from the spot valuation, the artificial spread between the futures price and spot price collapses.
3. **Backtest Data Integrity:** Quantitative research pipelines must anchor dividend adjustments strictly to the **Ex-Date**. Aligning data to payment or record dates introduces spurious arbitrage signals and perceived tracking errors between synthetic futures and spot series.

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